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Chemical structure characterization of polysaccharides using diffusion ordered NMR spectroscopy (DOSY)

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The potential of DOSY NMR spectroscopy to distinguish the linkage pattern of chemically related polysaccharides was evaluated using β-glucans isolated from yeast (*Saccharomyces cerevisiae*) and mushroom (*Pleurotus eryngii*). Laminarin from *Laminaria digitata* was included for chemical shift comparison. Characterization through methylation and 1D/2D NMR analysis showed that all the samples were constituted by →3)-Glc*p*-(1→; →3,6)-Glc*p*-(1→; Glc*p*-(1→ and →6)-Glc*p*-(1→ linkages. The results obtained allowed the identification of the well-known chemical structure of laminarin. Moreover, DOSY demonstrated that the units →3)-Glc*p*-(1→ and →6)-Glc*p*-(1→ from yeast β-glucans, presented the same diffusion time. For the mushrooms β-glucans, the diffusion time of these units were different, confirming that they belong to distinct polysaccharides. The yeast and the mushroom polysaccharide samples presented the same NMR correlations, but after DOSY analysis, different structures could be proposed. Therefore, DOSY NMR spectroscopy could be a tool for the identification of different linkage patterns of polysaccharides belonging to the same group and may be a useful contribution to the chemical structure and biological activity correlation studies of such structures.

1. Introduction

Polysaccharides are widely distributed in plants, fungi, animals, and microbes (;). The study of isolated and purified polysaccharides allowed to identify those polymers as potential antitumor and immunostimulant molecules. Their activities rely on the polysaccharide chemical features, such as monosaccharide composition, linkage pattern, anomeric configuration, branching degree, content of other functional groups (for instance, sulfate, carboxyl, acetyl) and molar mass (R. ;).

Therefore, the biological evaluation of polysaccharides first requires a detailed study of their chemical structure. For that purpose, classical techniques of carbohydrate chemistry have been applied, such as methylation analyses and 1D/2D nuclear magnetic resonance (NMR) experiments, allowing the identification of the distinct units that built up these polymers (;). However, due to the similarity that distinct polysaccharides may present, some of them cannot be separated by chromatography or other fractionation methods and can lead to ambiguous interpretation of the data (;). Therefore, the introduction of advanced techniques may be required for a proper characterization of polysaccharides.

Diffusion ordered NMR spectroscopy (DOSY) allows to determine the diffusion of molecules in solution through the application of gradient fields. A pulse sequence of DOSY NMR contains at least a pair of gradient pulses, the first one encodes the diffusion information while the second one decodes it, allowing a spatially label of the molecule (; ;). After the decoded pulse has been applied, the signal is detected. If the molecule has diffused, the intensity of the signal detected on the second pulse decreases in relation to the first one. For the molecules that present a low diffusion rate, the intensity of the signal is less affected. This pair of pulses is repeated stepwise throughout the experiment with increasing gradient strengths which results in a loss of the peak intensity and allows for obtaining a decay function curve (; ;). The diffusion rate can be obtained from this curve, given by the following equation:

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$$I = I_0 e^{-D\gamma^2 g^2 \delta^2 \left(\Delta - \frac{\delta}{3}\right)} \quad (1)$$

where I is the observed intensity, I_0 is the reference intensity (unattenuated signal intensity), D is the diffusion coefficient, γ is the gyromagnetic ratio of the observed nucleus, g = gradient strength, which increases at each step of the experiment, δ = length of the gradient, meaning the period of time the pulse lasts and Δ = diffusion time, meaning the time between the encode and decode pulses.

Regarding its applications, DOSY allows the determination of M_w, analysis of mixtures, detection of impurities, study of aggregation and interactions between different molecules (; ; ; ;). By using quantitative DOSY (qNMR), for instance, it is possible to quantify the amounts of glucose and sucrose of beverages, making this technique promising in food analysis (;). In another study, DOSY allowed to infer that β-glucans form a complex with glycogen through covalent bonds in the cell-walls of some *Candida* sp., and therefore, can also be a tool to understand the cell wall architecture of pathogenic fungi ( ).

In addition to all these applications, DOSY could also be a potential assignment aid, since it may allow for identification of different molecules in mixtures according to their diffusion coefficients ( ). DOSY has already been employed for the chemical characterization of β-glucans ( ). It also allowed to distinguish a series of *N*-acetyl-chitooligosaccharides presenting different degrees of polymerization (DP) (P.). Despite its potential, DOSY is still an undervalued technique in the carbohydrate chemistry field (;).

β-D-Glucan is a class of polysaccharides recognized as pathogen associated molecular pattern (PAMP) molecules, and therefore with potential immunostimulant activity (R.). The main β-D-glucan found in mushrooms and algae are composed of 1→3 linked β-D-glucopyranose with glucose residues attached to some of the units at the *O*-6 position (;). Differently, yeasts biosynthesize 3-linked β-D-glucan occasionally branched at *O*-6 by 1→6 and/or 1→3 linked β-D-glucopyranose side chains (;). In addition, 3-linked β-D-glucans with different side chains, as well as 6-linked β-D-glucans with or without branching at *O*-3 by a single glucose residue or by a long side chain, can also be present in yeasts and/or mushrooms (; ;).

In the present work, we evaluated the potentials and limitations of using DOSY NMR spectroscopy to distinguish distinct linkage pattern of structurally related polysaccharides. Our hypothesis was that DOSY is a potential technique to distinguish the linkage pattern of samples constituted by similar units. β-glucans were selected in the present work considering the fact that although being built up only by glucose residues, they comprise a wide range of structurally diverse molecules. First, the NMR assignments of β-glucans from three different sources, laminarin from the brown seaweed *Laminaria digitata*, and water-soluble β-glucans from the yeast *Saccharomyces cerevisiae* and from the mushroom *Pleurotus eryngii* were compared. Then, the samples containing mixtures of β-glucans were analyzed using diffusion ordered spectroscopy (DOSY).

2. Materials and methods

2.1. Materials

Laminarin from *L. digitata* was purchased from Sigma-Aldrich (Norway). The yeast β-glucan (Wellmune, Biothera, Eagan, MN, USA) was a gift from Immitec (Tønsberg, Norway). Laminarin and the yeast β-glucan were identified in the present work as **LMN** and **YBGS**, respectively. Fraction **PEA2-SP** from *Pleurotus eryngii* was previously obtained and characterized ( ). Briefly, the dried and milled fruiting bodies of the mushroom were sequentially extracted with dichloromethane (DCM), ethanol (EtOH), water and NaOH 0.05 M to remove other types of molecules that may be co-extracted with the β-glucan. Then, the obtained residue was extracted using NaOH 0.5 M, giving the fraction **PEA2**. This crude extract was fractionated according to its solubility in water and EtOH, giving rise to **PEA2-SP**, a fraction soluble in water and insoluble in EtOH.

2.2. Determination of carbohydrate content and monosaccharide compositions

The total carbohydrate content of the samples (1 mg/mL in water) was determined using the phenol/sulfuric acid assay with glucose as reference (10–50 μg/mL) and absorbance measurement at 490 nm ( ). The experiments were performed in quadruplicate. Results were expressed as average (in %) ± standard deviation (SD).

To determine the monosaccharide composition, approximately 1 mg of each sample were submitted to methanolysis reaction followed by TMS derivatization, as previously described ( ). The derivatives were analyzed by capillary gas chromatography coupled to a Focus GC (Thermo Scientific, Waltham, MA, USA). The column used was a Restek-Rtx-5 silica column (30 m, i.d. 0.32 mm, 0.25 μm film thickness). Helium was used as the carrier gas at constant pressure mode. The injector and the detector temperatures were 250 and 300 °C, respectively. The initial temperature of the column was 140 °C and after the injection (split injection of 1/10) it was increased by 1 °C/min to 150 °C, held for 3 min, increased by 3 °C/min to 170 °C, held for 5 min, increased by 15 °C/min to 310 °C, and held for 3 min. Chromatograms were analyzed using the Chromelion v.6.80 (Dionex Corporation, Sunnyvale, CA, USA) software.

2.3. Weight-average molar mass (M_w) determination

The M_w, the dispersion of M_w (Đ) and the recovery from a SEC column were determined by *SEC-MALLS-RI* using a Shimadzu LC-20HPLC system connected to a Dawn Helios +8 eight angle light scattering detector (MALLS) and an Optilab T-rex RI detector (Wyatt, California, USA). The system was equipped with a serially connected guard-column (Tosoh PWXL) and two size-exclusion columns (Tosoh TSK-gel G6000 PWXL and G6000PWXL). The injection loop delivered was 100 μL and the flow rate was 0.5 mL/min. Samples were dissolved in the eluent (0.1 M sodium nitrate/0.02 % azide solution) as described by . The refractive index increment (dn/dc) was considered to be 0.146 mL/g. The raw data were collected and processed with Astra software version 6 (Wyatt, USA).

2.4. Linkage pattern determination

The linkage pattern determination was performed though methylation analysis as previously described (;). Briefly, carbohydrate samples (~1 mg) were methylated with methyl iodide in dimethylsulfoxide, followed by the addition of NaOH. The *per-O*-methylated polysaccharides were first hydrolyzed with formic acid 90 % (500 μL, 80 °C, 60 min) and then with trifluoracetic acid 2.5 M (500 μL, 2 h, 100 °C). After hydrolysis, samples were reduced with a solution of sodium borodeuteride 0.5 M ($NaBD_4$) in ammonium hydroxide 2 M, (500 μL, 60 °C, 60 min) and acetylated by adding 200 μL of 1-methylimidazole and acetic anhydride (10 min, room temperature). The partially *O*-methylated alditol acetates were finally extracted with DCM and analyzed on a GC–MS-QP2010 (Shimadzu Corporation) equipped with a Restek Rxi-5MS silica column (30 m, i.d.

0.25 mm, 0.25 μm film thickness) using helium as carrier gas and split injection at constant pressure mode. The initial gas flow was 1 mL/min with interfacial temperature of 280 °C. The temperature of the column was 80 °C. Five minutes after the injection it was allowed to increase by 10 °C/min to 140 °C, then by 4 °C/min to 210 °C, then by 20 °C/min to 310 °C, and held for 4 min. The ion source temperature was 200 °C. Spectra were analyzed using GC–MS solution v.2.10 (Shimadzu Corporation) software.

2.5. Nuclear magnetic resonance (NMR) analysis

NMR analyses were performed on a Bruker Avance III HD 800 MHz spectrometer equipped with 5-mm cryogenic CP-TCI z-gradient probe (Bruker, Billerica, MA, USA) using deuterated dimethyl sulfoxide (DMSO-d_6 99.5 %, Sigma-Aldrich) as a solvent. Samples (approximately 10 mg) were dissolved in 500 μL of DMSO-d_6. The following experiments were performed at 55 °C with or without CW presaturation for suppression of water signals: ^{1}H (with continuous wave presaturation, pulse program "zgpr", TD = 131,072, SI = 131,072, NS = 16, SW = 18 ppm, D1 = 2 s), ^{13}C (zrestse.dp.jcm800, TD = 65,536, NS = 24,579, DS = 4, D1 = 2 s, P1 = 12 μs, CNSTO = 60°, P0 = P1*CNST0/90), HSQC (awhsqcedetgpsisp2.3 − 135pr, TD(F1) = 256, TD(F2) = 4096, NS = 16, DS = 16, D1 = 1.5 s, CNST2 = 145 Hz), selective HSQC (awshsqc135pr, TD(F1) = 128, TD(F2) = 2048, NS = 128, O2P = 100, spectral width (F1) = 20 ppm, spectral width(F2) = 10 ppm, DS = 32, D1 = 2 s, CNST2 = 145), COSY (cosygpprqf, TD(F1) = 256, TD(F2) = 2048, NS = 16, DS = 16, D1 = 2 s), TOCSY (dipsi2phpr, TD(F1) = 512, TD(F2) = 2048, NS = 16, DS = 16, D1 = 1.5 s, D9 = 80 ms), HMBC (awhmbcpr, TD(F1) = 256, TD(F2) = 4096, NS = 64, DS = 16, D1 = 1.5 s, CNST2 = 145 Hz, CNST13 = 8 Hz, D16 = 0.2 ms, D6 = 62.5 ms and awshmbcctetgpl2nd.m, TD(F1) = 128, TD(F2) = 2048, NS = 128, DS = 32, D1 = 1.5 s, CNST6 = 120 Hz, CNST7 = 170, CNST13 = 8 Hz), NOESY (awnoesygppr, TD(F1) = 128, TD(F2) = 4096, NS = 16, D1 = 2 s, D8 = 250 ms), ROESY (roesyph, TD(F1) = 256, TD(F2) = 4096, NS = 32, DS = 32, D1 = 1.5 s, p15 = 250 ms), H2BC (h2bcetgpl3, TD(F1) = 160, TD(F2) = 2048, NS = 64, DS = 16, D1 = 1.5 s, CNST6 = 120 Hz, CNST7 = 170 Hz) and HSQC-TOCSY (awhsqcdietgpsisppr, TD(F1) = 256, TD(F2) = 2048, NS = 92, D1 = 1.5 s, DS = 16, D9 = 80 ms). Spectra were processed and analyzed using TopSpin 3.6.2 and calibrated at 2.50 ppm for ^{1}H and at 39.40 ppm for ^{13}C.

2.6. Diffusion ordered spectroscopy (DOSY)

DOSY was recorded on a Bruker Avance III HD 800 MHz spectrometer (item 2.5). The pulse program used was a STE with bipolar gradient pulse pair, 1 spoil gradient (stebpgp1s) as follows: TD(F1) = 32, TD(F2) = 65,536, NS = 16, DS = 8 and D1 = 1 s. Spectra were recorded using ½ δ (also referred as p30) of 2000 μs, Δ (also referred as d20) of 0.1 s and gradient strength (gpz6) of 95–2 %. The recovery delay (d16) was 0.2 ms. The area of selected peaks was integrated and the method for error estimation was fit using arbitrary uncertainties with 95 % of confidence level. Spectra were analyzed using the dynamics center 2.8.4 software.

3. Results

3.1. Carbohydrate content and monosaccharide composition of the β-glucans samples

The carbohydrate content, as well as the monosaccharide composition of **LMN** and **YBGS** were determined, and the results are presented in .

Data from **PEA2-SP**, a β-glucan fraction isolated from *Pleurotus eryngii*, ( ) were included in for comparison. The carbohydrate content of all samples was >90 %. The monosaccharide composition showed that glucose was the major component in all fractions, ranging from 87 to 99 mol%.

Table 1
Carbohydrate content, monosaccharide composition and methylation analysis of the β-glucans samples.

Analyses	LMN	YBGS	PEA2-SP
Carbohydr. content (w/w%)	96 ± 4	96 ± 4	90 ± 4
Monosaccharide composition (w/w%)			
Fuc	2	n.d.	n.d.
Xyl	<1	n.d.	1
Man	n.d.	1	4
Gal	<1	n.d.	5
Glc	97	99	87
GlcA	n.d.	n.d.	2
GalA	n.d.	n.d.	1
Methylation analysis (% out of total)			
Glc*p*-(1→	9	7	15
→3)-Glc*p*-(1→	82	76	20
→3,6)-Glc*p*-(1→	7	5	12
→6)-Glc*p*-(1→	2	11	34
→4)-Glc*p*-(1→	n.d.	n.d.	9
→4,6)-Glc*p*-(1→	n.d.	n.d.	1

[a] , included for comparison. n.d. = not detected.

3.2. M_w distribution of the β-glucans samples

The M_w, Đ and recovery from SEC column obtained for **LMN** and **YBGS** are presented in and the elution profile in Fig. S1. **LMN** showed a homogeneous profile with M_w of 5 kDa (Đ 1.002). Differently, **YBGS** showed three different peaks, with Đ values of each peak varying from 1.1 to 1.5, indicating that this fraction was highly heterogeneous regarding the M_w distribution. The main peak was constituted of a pool of polysaccharides with an estimated M_w of 8 kDa. In addition, populations of polysaccharides of M_w 35 and 240 kDa were also identified in **YBGS**.

The M_w and recovery from SEC column for **PEA2-SP** was previously determined and were included in for comparison. Briefly, **PEA2-SP** presents an apparent M_w of 36 kDa. Populations with M_w higher than 2120 kDa were also identified. The higher M_w populations present both in **YBGS** and **PEA2-SP** are attributed to the presence of aggregates in aqueous solutions.

3.3. Structural elucidation of the β-glucans samples

To identify how the differences in the chemical structure of **LMN**, **YBGS** and **PEA2-SP** can be accessed when submitted to the classical methods in carbohydrate chemistry, samples were first analyzed through methylation analysis and then NMR.

The ^{1}H and ^{13}C NMR assignments were determined using HSQC, COSY, TOCSY, HSQC-TOCSY, NOESY, ROESY, HMBC and H2BC and were confirmed using the literature (; ; ; ; ; ; ; ; ;). The H2BC spectra were interpretated by overlapping them with the HSQC spectrum (

Table 2
Apparent M_w distribution and conformational parameters of the β-glucans samples.

Peak	Sample	M_w (kDa)	Đ (M_w/M_n)	Recovery (%)[b]
1	LMN	5	1.002	92
	YBGS	8	1.1	36
	PEA2-SP	36	1.8	69
2	YBGS	35	1.1	28
	PEA2-SP	2120	1.3	15
3	YBGS	240	1.5	19
	PEA2-SP	2425	1.002	6

[a] , included for comparison. Total recovery for **LMN**, **YBGS** and **PEA2-SP** was 92, 83 and 90 %, respectively.

). The NMR assignments are presented in .

3.3.1. LMN

Methylation analysis of **LMN** ( ) showed that this fraction is composed mainly of →3)-Glc*p*-(1→ (82 %) and lower amounts of →3,6)-Glc*p*-1(→ (7 %), corresponding to the branching points. In accordance with the percentage of branching points, 9 % of Glc*p*-(1→ was identified, corresponding to glucose as non-reducing terminal (NRT). In addition, small amounts 6-linked glucose units were identified (2 %).

The anomeric C/H correlations obtained through selective HSQC are presented in A. The HSQC spectrum of **LMN** is presented in Fig. S2.

In the anomeric region of the selective HSQC spectrum of **LMN**, four cross-peaks could be identified: at 102.7/4.52–4.53, 102.7/4.22, 103.5/4.38 and 102.9/4.30 ppm (A).

The main anomeric correlation was 102.7/4.52–4.53 ppm, attributed to the *C*-1/*H*-1 of the 1➔3-linked-β-D-glucose (unit A). The HMBC cross peaks (A) appearing from the linked *C*-3/*H*-3 (86.0/3.48 ppm) correlating with the anomeric signals at 102.7/4.52 ppm are in accordance with this assignment.

Additionally, literature reports that the *C*-1/*H*-1 correlations of the →3,6)-Glc*p*-1(→ (unit B) may be overlapped with those from unit A (; ; ; ;). The correlation at 74.5/3.51 ppm was attributed to the *C*-5/*H*-5 of unit B according to the literature ( ). HSQC-TOCSY cross-peaks between 74.5 ppm and 4.10;3.53 ppm (*H*-6), 3.30 ppm (*H*-4) and 3.34 ppm (*H*-2) are in accordance with this assignment (B).

Another main correlation in the HSQC spectrum of **LMN** was observed at 102.7/4.22 ppm (unit C) and was attributed to the *C*-1/*H*-1 of Glc*p*-(1→ attached to *O*-6 of unit B. The HMBC cross-peaks between 102.7/4.22 ppm (*C*-1/*H*-1, unit C) and 68.2/4.10; 3.53 ppm (*C*-6/*H*-6, unit B) supports that unit C is linked to the *O*-6 of unit B.

The presence of *C*-1/*H*-1 correlations of another NRT was identified in the HSQC spectrum of **LMN** at 103.5/4.38 ppm (unit D) ( ). A HMBC cross-peak between this anomeric correlation and a linked *C*-3/*H*-3 at 86.5/3.44 ppm could be observed. Unit D was attributed to the NRT of the main chain and the *C*-3/*H*-3 cross-peak at 86.5/3.44 ppm to the second monosaccharide unit after the NRT, which was named as A'. The presence of unit D and A' agrees with the relatively low M_w determined for this fraction (5 kDa). Unit A' could also correspond to correlations of 3-linked-glucose units adjacent to branch points (; ;).

The remaining assignments of units A-D were determined using COSY, TOCSY, HMBC, HSQC-TOCSY, H2BC, NOESY, ROESY as described in and are in accordance with the literature.

The anomeric HSQC correlation at 102.9/4.30 ppm was ascribed as the *C*-1/*H*-1 of →3)-Glc*p*-(1→ (Unit E) linked to a primary hydroxyl of a mannitol residue (Unit M) in the reducing terminal (RT). This was confirmed through the HMBC cross-peaks from 102.9/4.30 ppm (*C*-1/*H*-1, Unit E) with 72.1/4.03 and 3.51 ppm (M1) and through the presence of HSQC correlations from the terminal primary hydroxyl group of mannitol at 63.5/3.63; 3.42 ppm (M6) ( ). The remaining assignments of unit E were ascribed using COSY, TOCSY, HSQC-TOCSY, H2BC and ROESY as represented in .

From the methylation and NMR analysis it was possible to identify that **LMN** is a 3-linked β-D-glucan branched at *O*-6 by one single glucose residue in an approximate ratio of 1:10 and presenting a mannitol group attached to at least some of the RT, as described in literature (J. ; ;). Laminarins can have either mannitol or glucitol group, being classified in M chains and G chains attached to their RT. For laminarin from *Laminaria* sp. it has been estimated the presence of 40–75 % of M chains (J. ; ;). Although glucitol correlation could not be detected, low percentage of these residues as the RT of **LMN** may still be present.

3.3.2. YBGS

The methylation analysis of **YBGS** ( ) showed the presence of →3)-Glc*p*-(1→ and →6)-Glc*p*-(1→ (76 and 11 %, respectively). In addition, 7 % of glucose as NRT, i.e. Glc*p*-(1→, were identified, which is in accordance with the presence of 5 % of branching points, ➔3,6)-Glc*p*-(1→. The structure of **YBGS** was further investigated by 1D/2D NMR analyses.

The HSQC spectrum of **YBGS** is presented in Fig. S3. In the anomeric region of the selective HSQC spectrum, the *C*-1/*H*-1 correlations of units A, C and D were identified (B), as described for **LMN**.

Furthermore, the HSQC spectrum of **YBGS** also presented four additional anomeric correlations: 103.0/4.27, 102.4/4.36, 103.7/4.37 and 103.1/~4.45 ppm (B). Minor HSQC correlations at 91.6/4.99 and 96.2/4.38 ppm were also present and were attributed to the RT of glucose in its α and β configuration, respectively. This is in accordance with the presence of low M_w polysaccharides (8 kDa), as indicated by the SEC-MALLS analysis ( ).

The HSQC correlation at 103.0/4.27 ppm was attributed to →6)-Glc*p*-(1→ (unit F) (; ;). HMBC correlations (A) between *C*-1/*H*-1 of unit F and a deshielded *C*-6/*H*-6 (68.3/4.00; 3.60 ppm) are in accordance with presence of 1→6 linked glucose units. The ^{1}H chemical shifts of this linked C-6/H-6 are different from those attributed to unit B and therefore, can be useful to distinguish between a 6-linked glucose chain and a branch point when the remaining chemical shifts present low intensity and/or are overlapped with correlations arising from other units. The correlations of the ring region of unit F were confirmed through COSY, TOCSY, HSQC-TOCSY, NOESY and ROESY correlations and with literature data ( ). The →6)-Glc*p*-(1→ units in yeast β-glucan samples have been ascribed to both side and main chains (;).

The anomeric correlation at 102.4/4.36 ppm was designated unit G. In the HSQC-TOCSY spectrum (B), this signal correlated with a *C*-6 at 60.6 ppm and a deshielded *C*-3 at 87.2 ppm (*H*-3 at 3.37 ppm according to TOCSY), indicating that unit G is 1➔3-linked glucose residues in a different environment than unit A. C-4/H-4 (68.0/3.32 ppm) and C-5/H-5 (76.0/3.27 ppm) cross-peaks were also assigned through HSQC-TOCSY. In the HMBC spectrum of **YBGS**, a cross-peak between the anomeric 102.4/4.36 ppm (Unit G) and a 68.2/4.00; 3.60 ppm methylene suggested that unit G is attached to the *O*-6 of the adjacent unit (non-identified branch point).

The correlation at 103.7/4.37 ppm (*C*-1/*H*-1 unit H) could also be attributed to a NRT attached to the *O*-3 of the adjacent unit ( ). In our previous work, the same correlation was found for the β-glucans from *P. eryngii*, and it was attributed to the NRT attached to the *O*-3 of a 1➔6-linked β-D-glucan.

In addition, HMBC cross-peaks were observed between 103.1/~4.45 ppm with *C-3/H-3* of RT glucose in its α (84.4/3.63 ppm) and β (87.2/3.39 ppm) configuration. Therefore, the correlation at 103.1/~4.45 ppm was ascribed to the C-1/H-1 of the unit adjacent to the RT.

Although methylation analysis has showed the presence of branch points, such could not be detected in the HSQC NMR spectrum of **YBGS**. This could be due to the presence of different types of branching points and the possibility that they overlapped with the correlation of other units. Therefore, six different units were identified in **YBGS** A, C, D, F, G and H. Different β-glucans have already been proposed in literature, such as 3-linked β-glucan branched at *O*-6 by a 6-linked and/or 3-linked glucose side chain, 6-linked β-glucan branched at *O*-3 by a 3-linked glucose side chain and 6-linked β-glucan branched at *O*-3 by one glucose residue (;). From the results obtained in the present work, all these types of β-glucans could be present in **YBGS**.

Table 3
NMR assignments of the units identified in samples LMN, YBGS and PEA2-SP.

Monomer		^{1}H	^{13}C	COSY	TOCSY	HMBC	HSQC-TOCSY	H2BC	NOESY	ROESY
Unit A[1,2,3] →3)-Glc*p*-(1→	1	4.52–4.53	102.7	A2	A2, A3, A4/ A5	A3	A1, A2, A3, A4, A5, A6	A2	A3, A5	A3, A5
REF: a, b, c, d, e, f, i	2	3.29	72.5	A3	A3, A1	A1, A3	A1, A2, A3, A4, A5, A6	A1, A3		
	3	3.48	86.0	A4	A4/A5, A2, A6	A1, A2, A4	A1, A2, A3, A4, A5, A6	A2, A4	A1	A5
	4	3.25	68.2		A1, A3, A6	A3, A5	A1, A2, A3, A4, A5, A6	A3, A5		
	5	3.27	76.1		A1, A3, A6	A4, A6	A1, A2, A3, A4, A5, A6	A4, A6		A3
	6	3.70;3.46	60.6		A3, A4	A5	A1, A2, A3, A4, A5, A6	A5		
Unit A'[1,2]	1	4.52–4.53	102.7			A3		A'2		
→3)-Glc*p*-(1→	2	3.29	72.5			A'1, A'3		A'1, A'3		
REF: d, f, I Adjacent to unit D or to a branch point	3	3.44	86.5			A'2, A'4		A'2, A'4		
	4	3.25	68.2			A'1, A'3, A'5		A'3, A'5		
	5	3.27	76.1			A'4, A'6				
	6	3.70;3.46	60.6			A'5				
Unit B[1,3,]	1	4.52–4.53	102.7							
→3,6)-Glc*p*-(1→	2	3.34	72.2				B5, B6			
REF: a, c, d, e, f, i	3	3.50	85.8							
Branch point when the main chain is a 3-	4	3.30	68.8							
linked β-glucan	5	3.51	74.5			^{3}B6	B6, B4, B2			
	6	4.10; 3.53	68.2			C1, B5				B4
Unit C[1,2,3] Glc*p*-(1→	1	4.22	102.7	C2	C2, C3, C4, C5	B6	C1, C2, C3/C5, C4, C6	C2		C3, C5
REF: a, c, d, e, f	2	3.02	73.5	C3	C4, C3, C1	C1	C1, C2, C3/C5, C4, C6	C1, C3		
NRT attached to the *O*-6 of the adjacent unit	3	3.19	76.1				C1, C2, C3/C5, C4, C6	C2, C4		
	4	3.09	69.9		C3, C6		C1, C2, C3/C5, C4, C6	C3, C5		
	5	3.12	76.4				C1, C2, C3/C5, C4, C6			
	6	3.70; 3.46	60.6							
Unit D[1,2,] Glc*p*-(1→	1	4.38	103.5	D2	D2, D3	A'3	D1, D2, D3/D5, D4	D2		
REF: g	2	3.09	73.5	D3	D1, D3		D1, D2, D3/D5, D4, D6	D1, D3		
NRT attached to the *O*-3 of the adjacent unit	3	3.22	76.0				D2, D3/D5, D4, D6			
	4	3.09	69.9							
	5	3.22	76.2							
	6	3.70; 3.46	60.9							
Unit E[1]	1	4.30	102.9	E2	E2, E3, E4	M1	D2, D3	E2		M1
→3)-Glc*p*-(1→	2	3.26	72.0	E3			D4, D5, D6	E1, E3		
	3	3.44	86.5	E4		E1, E5		E2, E4		
	4	3.20	68.2							
	5	3.12	76.3			E1				
	6	3.70, 3.46	60.7							
Unit F[2] →6)-Glc*p*-(1→	1	4.27	103.0	F2	F2, F3, F4, F5	F5, F6	F1, F2, F3/F5, F4, F6	F2	F5 F6	F1, F3, F6
REF: a, f, g, h	2	3.02	73.2	F3	F1, F3, F4, F5	F1, F3	F1, F2, F3/F5, F4, F6	F1, F3		
	3	3.18	76.1		F5, F1, F2	F2, F4	F1, F2, F4	F2, F4		F1
	4	3.14	69.8	F5	F1, F2, F5	F3, F5	F1, F2, F3/F5, F4, F6			
	5	3.30	75.3			F4, F6	F1, F6			
	6	4.00, 3.60	68.3			F1	F5	F5	F1 A3	

(*continued on next page*)

Table 3 (*continued*)

Monomer		^{1}H	^{13}C	COSY	TOCSY	HMBC	HSQC-TOCSY	H2BC	NOESY	ROESY
Unit G[2] →3)-Glc*p*-(1→6	1	4.36	102.4	G2		G2, 68.2	G1, G2, G3, G4, G5, G6		G3	*G3*
3-linked unit attached to the *O*-6 of the	2	3.23	72.1			G1, G3				
adjacent unit	3	3.37	87.2			G2, G4	G1, G2, G3, G4, G5, G6			
	4	3.32	68.0			G3, G5				
	5	3.27	76.0			G4, G6				
	6	3.46, 3.70	60.7			G5				
Unit H[2,3]	1	4.37	103.7	H2		I3, H2				
Glc*p*-(1→	2	3.09	73.6	H3		H1, H3				
REF: f, g	3	3.19	76.6							
NRT attached to the *O*-3 of the adjacent	4	–	–							
unit	5	–	–							
	6	–	–							
Unit I[3,]	1	4.36	102.4	I2	I2, I3, I4	I2		I2	I3	
→6,3)-Glc*p*-(1→	2	3.24	72.0			I1, I3				
REF: f	3	3.39	87.2		I1, I2	H1, I2, I4				
Branch point when the main chain is a 6-	4	3.31	68.0			I3				
linked β-glucan	5	3.30[f]	75.8[f]							
	6	4.10, 3.53	68.2							
Unit M[1]	1	4.03, 3.51	72.1			E1				
	6	3.63, 3.42	63.5							
Unit-RT →3)-Glc*p*-(1→ Adjacent to a RT	1	~4.45	103.1			α- and β-Glc-RT (3)				
α-Glc-RT	1	4.99	91.6							
	3	3.63	84.4							
β-Glc-RT	1	4.38	96.2							
	3	~3.39	87.2							

Solvent: DMSO-d_6, 55 °C, ^{1}H = 2.50 ppm and ^{13}C = 39.4 ppm.
1 = Present in **LMN**, 2 = Present in **YBGS** and 3 = Present in **PEA2-SP**.
References: a = ; b = ; c = ; d = ; e = ; f = ; g = ; h = , i = .
Unit B and I could also be present in **YBGS**, probably overlapped with units A and G, respectively.
** For **YBGS**, unit D could belong to either the main chain or to the side chains.
*** Observed only in **PEA2-SP**.
**** Observed only in **YBGS**.
***** HMBC cross-peaks appearing from 102.4/4.36 ppm with C-6 carbon at 68.2 ppm, suggests unit G is a 3-linked-glucose attached to the *O*-6 of the adjacent unit.

3.3.3. *PEA2-SP*

The chemical structure of fraction **PEA2-SP** has been previously characterized by our group using methanolysis to determine the monosaccharide composition, and methylation and NMR analysis to determine the linkage pattern. The results obtained are summarized below.

PEA2-SP consisted mainly of glucose (, 87 %), indicating the presence of glucans and other types of polysaccharides. The later may remain in β-glucans samples even after several steps of extraction and purification ( ). Regarding the linkage pattern, methylation analysis of **PEA2-SP** β-glucans showed the presence of 34 % of 6-linked and 20 % of 3-linked-glucose. In addition, 9 % of 4-linked glucose were detected ( ). The anomeric correlations between 98.0–103.0/4.90–5.30 ppm indicate the presence of monosaccharide units in α configuration ( ). The identification of 4-linked-glucose and monosaccharide units in α configuration are in accordance with the presence of starch. Low intensity HSQC correlations at 99.6/5.09 ppm suggested that small amounts 3-linked α-glucans may be present (;). However, the remaining correlations of α-(1→3)-glucans could not be identified, which suggests that if present, it is in a low concentration.

In accordance with the presence of mainly β-glucans, the HSQC NMR spectrum showed high intensity correlations at 102.0–106.0/4.40–4.80 ppm. Moreover, the assignments of units A, B, C, F and H (C, and S4), were identified as represented in . The correlation at 102.4/4.36 ppm has been assigned in literature as →6,3)-Glc*p*-(1→, (unit I) corresponding to the branch point of the 6-linked-β-D-glucan (;). The HMBC cross-peaks between I3 and H1 are in accordance with this attribution (A).

From these results, we suggested in our previous work that the *P. eryngii* mushroom biosynthesizes two different types of branched β-glucans. The first one is a (1→3)-β-glucan occasionally branched at *O*-6 by single glucose residues, which is in agreement with the presence of units A, B and C. The other one is a (1→6)-β-glucan occasionally branched at *O*-3 by single glucose residues (;) which is in accordance with the presence of units F, H and I. The same types of β-glucans in *P. eryngii* alkali extracts have already been described in literature, besides the presence of a linear 6-linked β-glucan (; ;).

However, even though both **PEA2-SP** and **YBGS** showed the presence of the same methylated derivatives and similar NMR correlations, different chemical structures are described in literature for the β-glucans

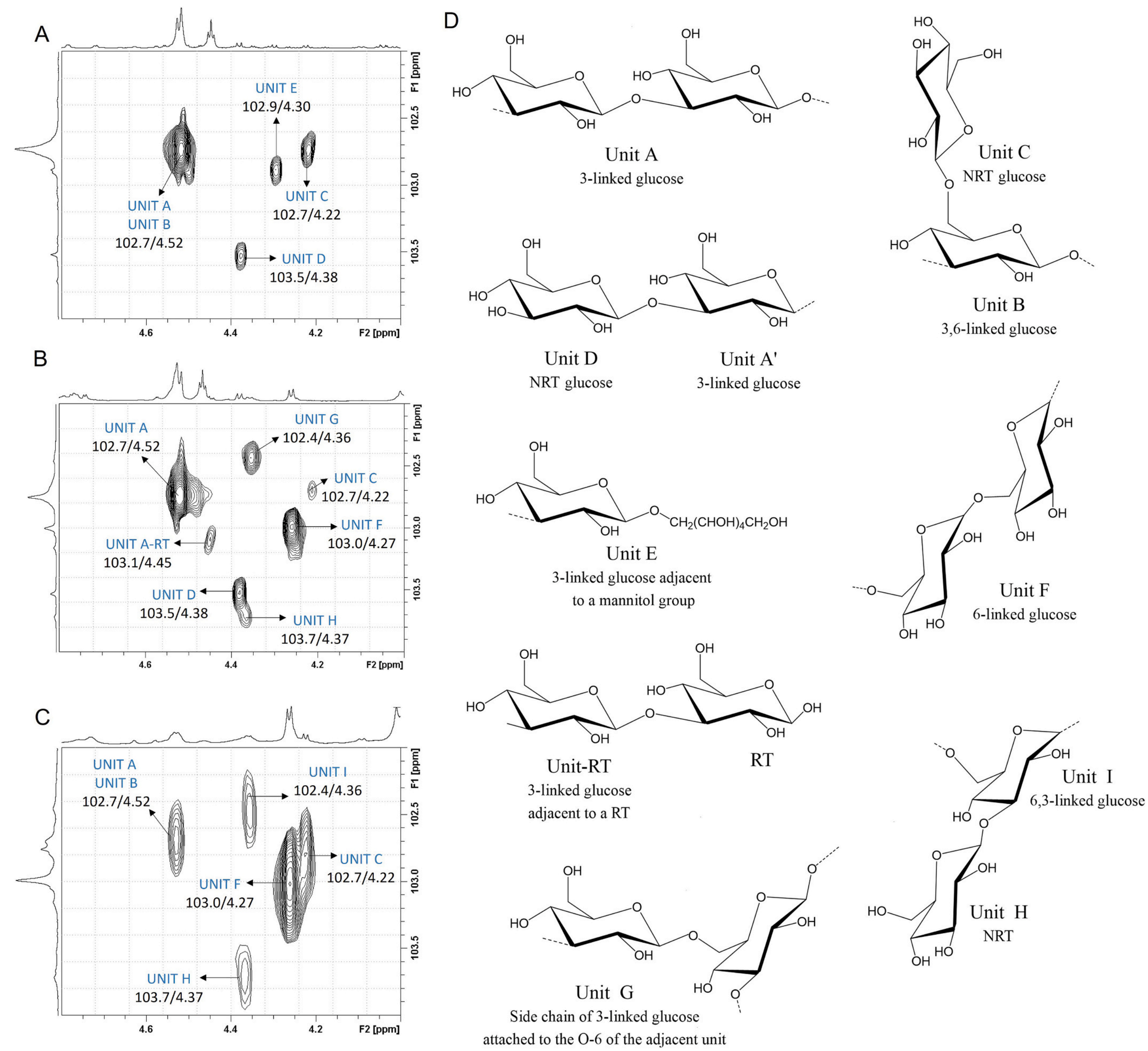

Fig. 1. Anomeric region of the HSQC NMR spectra of fractions **LMN** (A), **YBGS** (B) and **PEA2-SP** (C) and the schematic representation units identified in these samples (D).
Selective HSQC (O2P = 100, spectral width F1 = 20 ppm, spectral width F2 = 10 ppm).
NRT = non-reducing terminal and RT = reducing terminal.

obtained from these sources (; ;). This reinforces the need of more advanced techniques in the chemical characterization of polysaccharides. Considering this, the chemical structure of **PEA2-SP** was re-analyzed in the present work using DOSY NMR analysis.

3.4. DOSY NMR analysis

Methylation and 1D/2D NMR analysis showed that **YBGS** and **PEA2-SP** contain units →3)-Glc*p*-(1→; →3,6)-Glc*p*-(1→: Glc*p*-(1→ and →6)-Glc*p*-(1→. These units seem to belong to distinct polysaccharide chains. To verify the possibility of accessing the chemical structural differences between the mixtures of β-glucans from the different sources, samples were submitted to DOSY NMR analysis. For that purpose, only the anomeric chemical shifts were analyzed, since they are less overlapping than the chemical shifts arising from H-2/H-6 ( ). The result is presented in and .

For **YBGS**, the diffusion times of units A (4.53 ppm), F (4.27 ppm) and G (4.36 ppm) were determined. All the units presented similar diffusion time, suggesting that either they belong to the same polysaccharide chain, or to distinct polysaccharides with similar M_w.

For fraction **PEA2-SP**, the diffusion time of the anomeric correlations corresponding to units A/B (4.52 ppm), F (4.27 ppm), I (4.36 ppm) and C (4.22 ppm) were determined ( ). Two distinct diffusion time populations were observed, one built up of units A and C and a second type composed of units I and F. From this, it is possible to infer that for the mushroom's β-glucans, the units →3)-Glc*p*-(1→ and →6)-Glc*p*-(1→ belong to two distinct polysaccharide types.

Therefore, DOSY NMR analysis suggests that, although both samples contain the same units in different proportions, the linkage distribution

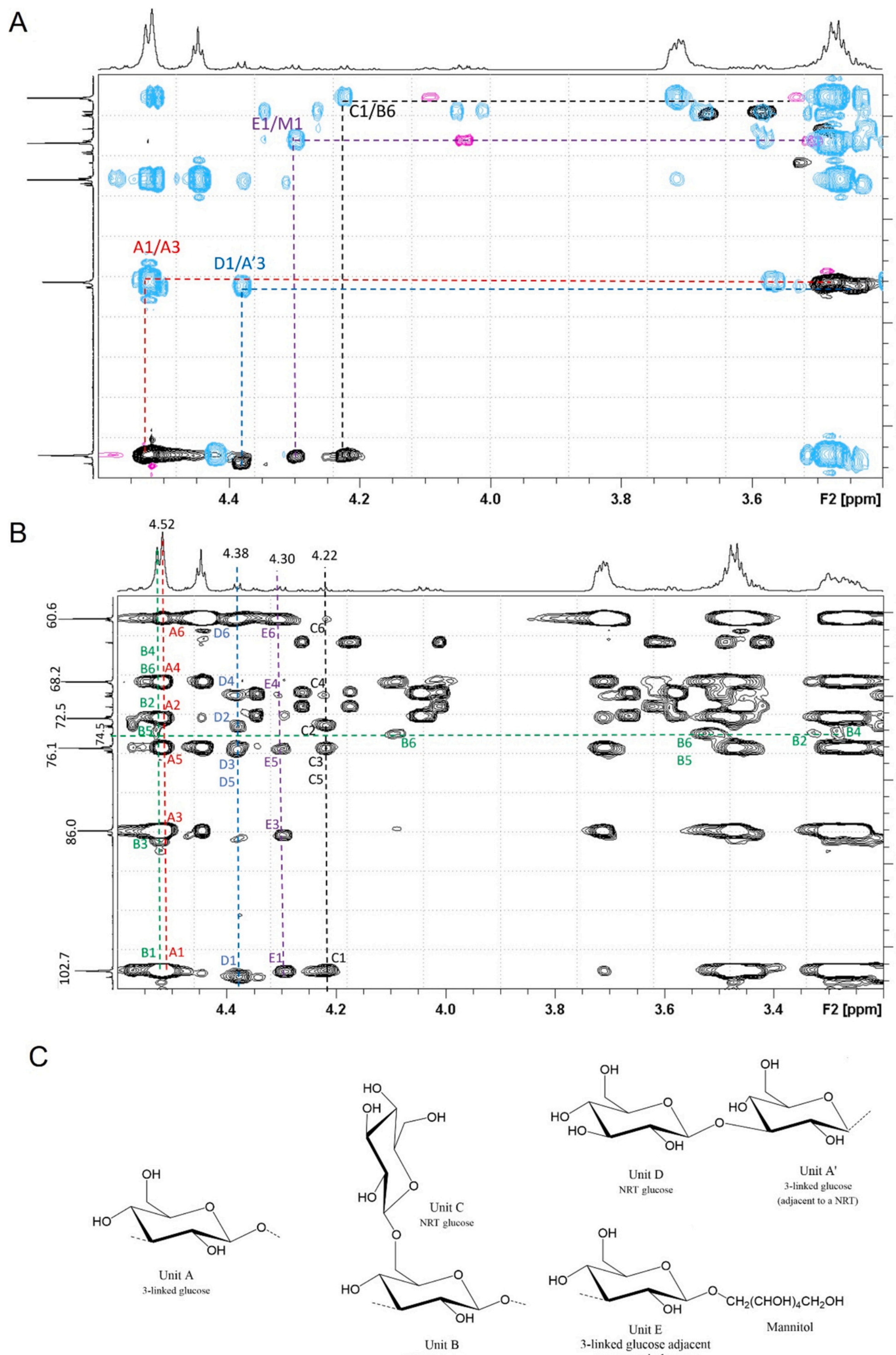

Fig. 2. NMR spectra of **LMN**. Superposition of HSQC (black) and HMBC (blue) spectra (A). HSQC-TOCSY spectrum (B). Units identified in **LMN** (C). The pink correlations in panel A correspond to the HSQC correlations arising from position 6. NRT = non-reducing terminal.

of the units are different. Different types of β-glucans could be proposed for **YBGS** and **PEA2-SP**, as described in the discussion session.

4. Discussion

4.1. Chemical structure characterization of the β-glucans samples

The chemical structure elucidation of polysaccharides is a mandatory step for the study of their biological properties. Studies on polysaccharides must contain information regarding their monosaccharide composition, type of linkages, anomeric configuration, ring configuration and molar mass. For that purpose, techniques such as gas and/or liquid chromatography, mass spectrometry and NMR are usually applied ( ).

Polysaccharides have been studied using well established protocols for the determination of their chemical structures, including

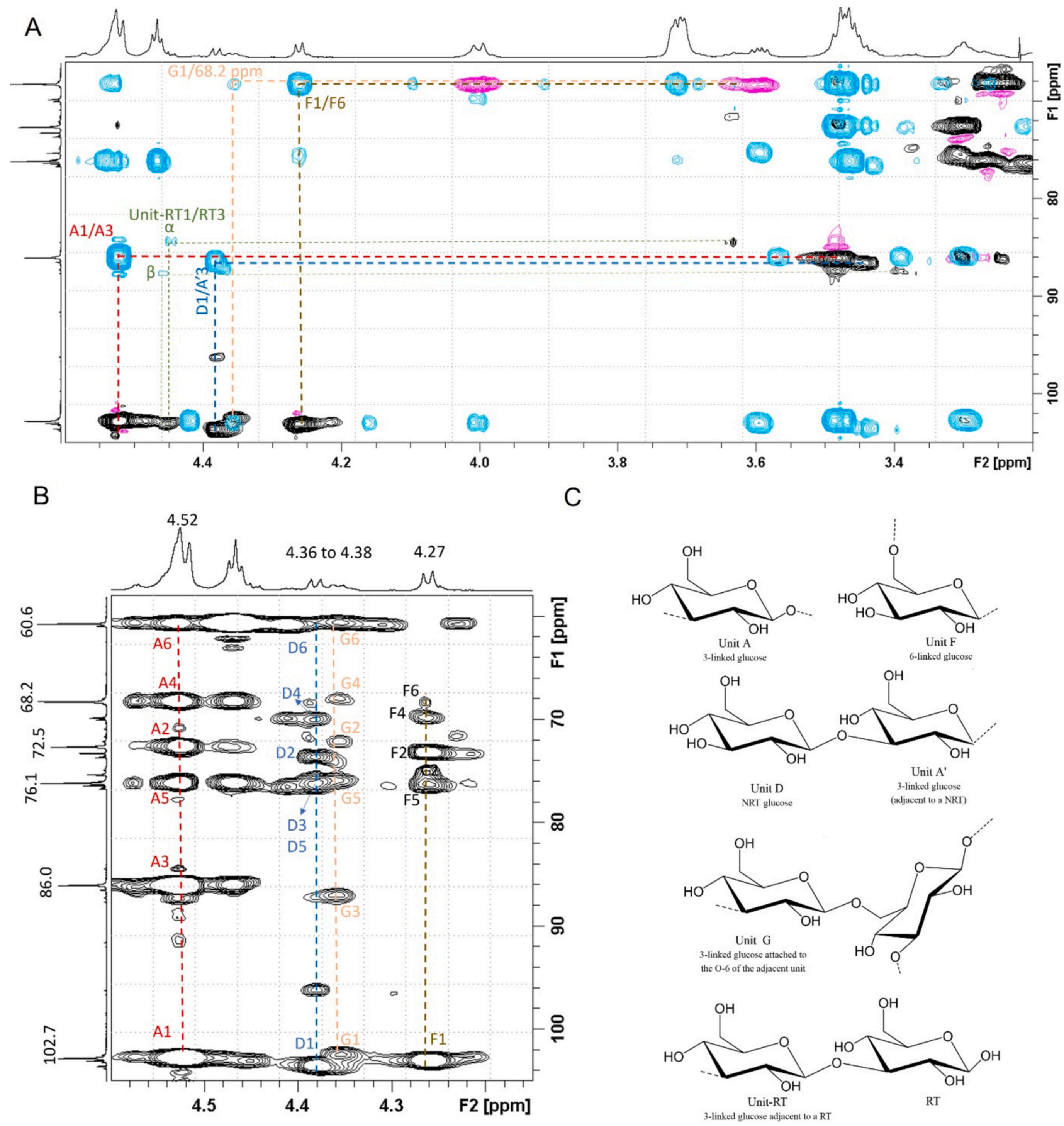

Fig. 3. NMR spectra of **YBGS**. Superposition of HSQC (black) and HMBC (blue) spectra (A). HSQC-TOCSY spectrum (B). Units identified in **YBGS** (C). The pink correlations in panel A correspond to the HSQC correlations arising from position 6. NRT = non-reducing terminal and RT = reducing terminal.

methylation and 1D/2D NMR analysis (;). However, due to the diversity of linkage patterns that those polymers may present, ambiguous conclusions may be obtained. Considering this, we have investigated the possibility of using DOSY NMR analysis as an auxiliary tool in the structure determination of complex carbohydrate samples, such as those containing mixtures of β-glucans. For that purpose, β-glucans samples obtained from yeast (*S. cerevisiae*) and mushroom (*P. eryngii*) were employed.

Laminarin was included for chemical shift comparison purposes. Using the classical methodologies in carbohydrate chemistry it was possible to determine the chemical structure of commercially available **LMN**, a homogeneous fraction regarding M_w and chemical structure. The HSQC anomeric region of this sample showed the presence of units A, A', B, C, D, E and M. The NMR assignments of the remaining *C/H* were resolved using COSY, TOCSY, HMBC, HSQC-TOCSY, H2BC, NOESY and ROESY, and are in accordance with the literature (; ; ; ; ; ;). The HMBC and HSQC-TOCSY spectra were useful to identify the presence of unit B, which seems to have overlapping *C/H* correlations with unit A at

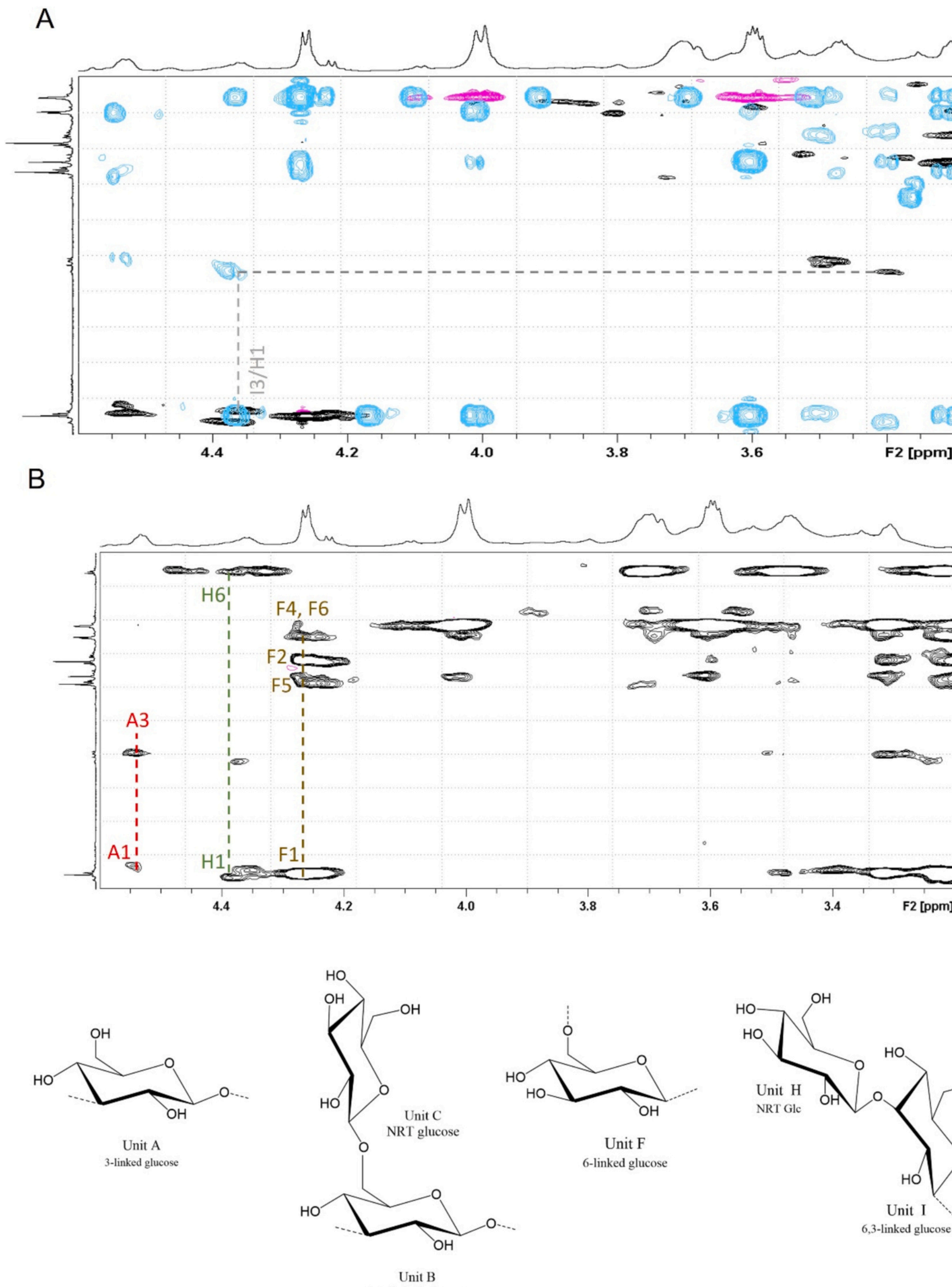

Fig. 4. NMR spectra of **PEA2-SP**. Superposition of HSQC (black) and HMBC (blue) spectra (A). Units identified in **PEA2-SP** (B). The pink correlations in panel A correspond to the HSQC correlations arising from position 6.

positions 1, 2, 3 and 4. Therefore, methylation and the above-mentioned NMR techniques were useful to determine the chemical structure of **LMN**, which showed to be a 1➔3-linked β-D-glucan branched at *O*-6 by single glucose units with a branch degree of 1:10. In addition, it was possible to identify the presence of a mannitol group attached to the RT, which according to the literature, may correspond to 40–75 % of the RT of laminarins. Glucitol as RT may also be present, although no NMR correlations could be identified, probably due to the low content of this residue and overlapping with the mannitol correlations. This result is in accordance with what is described in literature for laminarin, and the suggested chemical structure is represented in (J. ;).

Methylation analysis showed that the yeast (**YBGS**) and the mushroom (**PEA2-SP**) β-glucans samples are built up of the same main units in different proportions. **YBGS** contained almost 4 times more →3)-Glc*p*-(1→ and 3 times less →6)-Glc*p*-(1→ compared to **PEA2-SP**. The difference in proportion suggests the presence of distinct linkage pattern distribution, although it could not be identified using the 2D NMR analyses employed here. In order to identify how these units are linked together and access the differences between both samples, **YBGS** and

Table 4
Diffusion (m^2/s) of the anomeric hydrogens of samples YBGS and PEA2-SP.

Sample	Unit	Linkage pattern	δ ^{1}H (ppm) anomeric proton	Diffusion time (m^2/s)
YBGS	**A**	*→3)-Glcp-(1→*	4.53	$(9.8 \pm 0.4) \times 10^{-11}$
	F	*→6)-Glcp-(1→*	4.27	$(9.4 \pm 0.2) \times 10^{-11}$
	G	*→3)-Glcp-(1→6*	4.36	$(9.5 \pm 0.4) \times 10^{-11}$
PEA2-SP	**A/B**	*→3)-Glcp-(1→*	4.52	$(5.3 \pm 0.4) \times 10^{-11}$
	C	*Glcp-(1→*	4.22	$(5.2 \pm 0.3) \times 10^{-11}$
	F	*→6)-Glcp-(1→*	4.27	$(9.0 \pm 0.2) \times 10^{-11}$
	I	*→6,3)-Glcp-(1→*	4.36	$(9.3 \pm 0.8) \times 10^{-11}$

PEA2-SP were submitted to DOSY analysis.

For **YBGS**, DOSY showed that units F (6-linked), A (3-linked) and G (3-linked attached to the O-6 of the adjacent unit) presented similar diffusion time. Units A and G diffusing at the same time is in accordance with the presence of a 3-linked β-glucan branched at *O*-6 by a 3-linked glucose side chain, as described in literature (;) ( ). From the 2D NMR spectra, it could be inferred that unit D (NRT linked to the *O*-3 of the adjacent unit) could also belong to this structure and could be attributed to the NRT of the main and/or side chain.

In addition, the DOSY NMR spectrum of **YBGS** showed that for this pool of polysaccharides, Unit F also presents the same diffusion time as unit A. For yeast β-glucans, the presence of unit F has been attributed to the side chains linked to a 3-linked β-D-glucose and/or to a 6-linked-β-D-glucan branched at *O*-3 by one single glucose residue (; ;). The presence of unit H in NMR spectra of **YBGS** supports the presence of the latter. Although methylation analysis showed **YBGS** to contain 5 % of branching points, no correlations corresponding to these units could be identified, which could be attributed to the low content of branching points and overlapping of the signals. Therefore, it was not possible to infer if unit F is attached to unit A or it is another polysaccharide with similar M_w.

Although the main units in **YBGS** presented the same diffusion time,

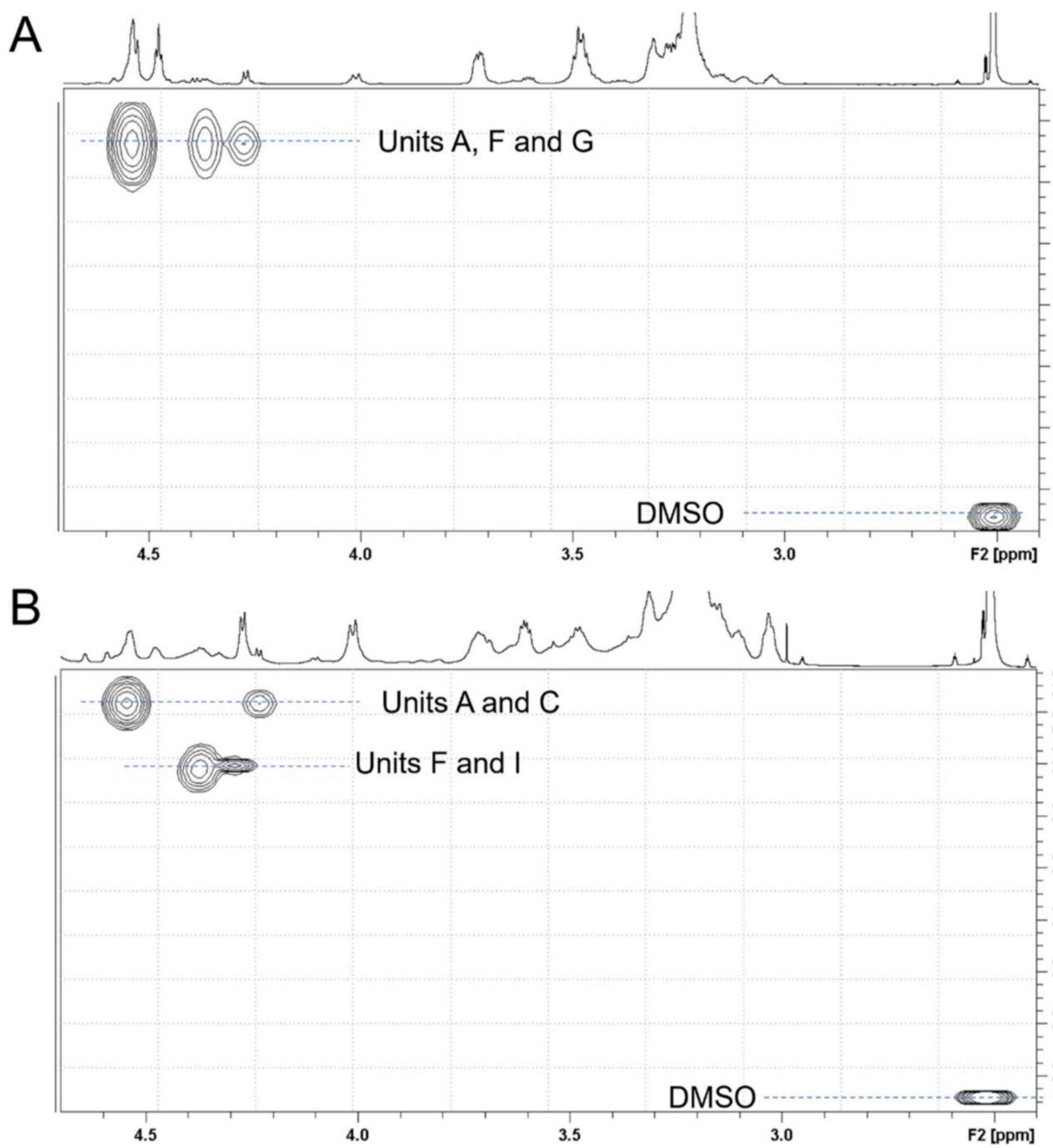

Fig. 5. DOSY NMR spectra of **YBGS** (A) and **PEA2-SP** (B).

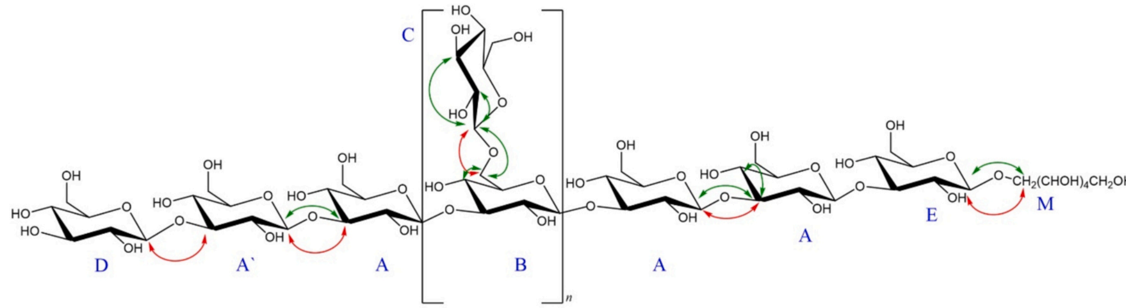

Fig. 6. Proposed chemical structure of the β-glucan present in **LMN**.
The arrows represent the HMBC (red) and ROESY (green) correlations. n = units B and C are distributed along the polysaccharide chain in an approximated ratio of 1:10. The figure represents laminarin from the M-series (mannitol attached to the RT), which may correspond to 40–75 %. Laminarin from the G-series (glucitol attached to the RT) may also be present in **LMN**.

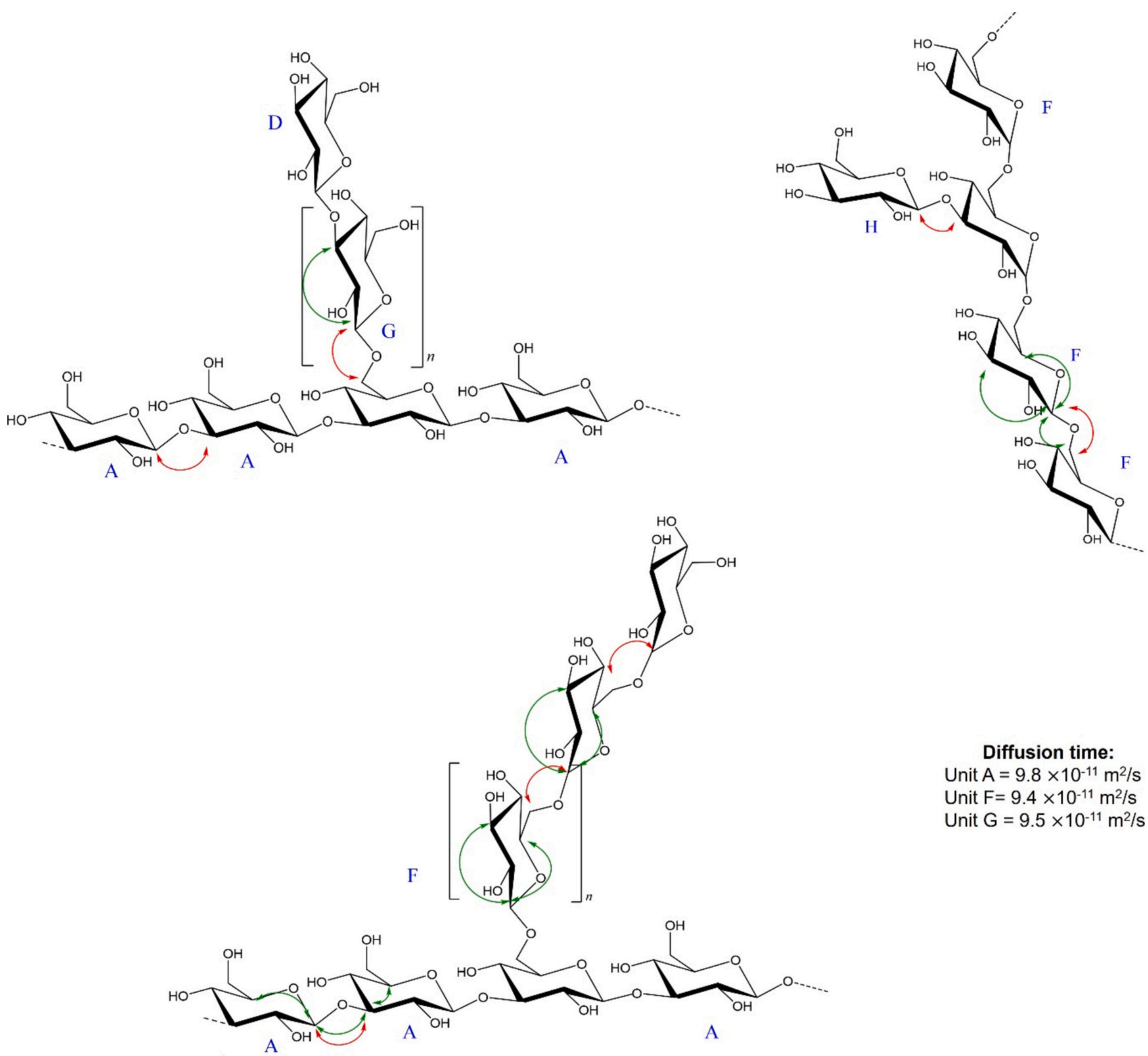

Fig. 7. Proposed chemical structures of the β-glucans that could be present in **YBGS**.
The arrows represent the HMBC (red) and ROESY (green) correlations.
All the chemical structures that could be proposed for **YBGS** after 2D NMR (HSQC, HMBC, H2BC, COSY, TOCSY, HSQC-TOCSY, NOESY and ROESY) and DOSY NMR analysis. The correlations of the branch point could not be detected due to the lower content of these units (5 %) and due to the fact that they are probably overlapped with unit B and I.

the SEC-MALLS-RI analysis detected a pool of polysaccharides of different M_w. This could suggest that in aqueous solutions, the different β-glucans adopt distinct macromolecular structures, leading to an apparent polydisperse M_w distribution. The types of structures that **YBGS** may contain are presented in .

For fraction **PEA2-SP**, DOSY showed that units F and A present different diffusion times, indicating that the 1→3 and 1→6-linked-glucose units belong to two distinct types of polysaccharides in the mushroom β-glucans. This is in accordance with the literature, which describes that *P. eryngii* synthesizes two different types of β-glucans: a 3-linked β-D-glucan branched at *O*-6 by single glucose residue, and a 6-linked β-D-glucan branched at *O*-3, also by single glucose units ( ) ( ).

Therefore, the results obtained here demonstrate that the characterization of complex carbohydrate mixtures can lead to misinterpretation of the data. Both **YBGS** and **PEA2-SP** presented similar HSQC correlations. After in-depth analyses, the different types of β-glucans could be proposed for the different samples, indicating that, although with some limitations, DOSY could be a tool in the chemical characterization of β-glucans and possibly for other types of polysaccharides.

4.2. DOSY parameters

In order to use DOSY in the chemical characterization of polysaccharides, some parameters such as the values of Δ, δ (Eq.) and the choice of the solvent should be taken into consideration for obtaining the best data quality possible.

According to the viscosity of the solvent and to the size of the polymer, the appropriate values of Δ (d20) and ½ δ (p30) may vary. For instance, for small molecules in deuterated chloroform ($CDCl_3$) the ideal values of Δ and 1/2δ have been reported to be 50 and 1 ms, respectively. Higher values are necessary for polymers in more viscous solvents ( ). In the present work, the Δ (d20) and 1/2δ (p30) values were 100 and 2 ms, respectively. The diffusion decay curves presented a linear profile (Figs. S5 and S6), indicating that these parameters were suitable for the β-glucan samples. Polymers with higher M_w may require higher p30 values. The d20 and p30 values are recommended by the manufacturer to be in the range of 50–100 and 1–3 ms, respectively. Other values may damage the hardware and/or the probe and therefore, should be selected with care.

Other critical parameter is related the choice of the solvent. Chloroform and other low viscosity solvents may undergo a convection process, due to temperature variation across the sample (;). For DMSO, convection currents are not expected, which allowed us to use a pulse sequence (stebpgp1s) without convection compensation ( ).

In addition, since the aim of this work was to detect distinct linkage patterns that a pool of chemically related polysaccharides may present, some other characteristics must be taken into consideration.

First, the polymer-polymer interactions should be minimized. This would allow to obtain distinct diffusion times for different polysaccharides if both polymers don't present the same M_w. Solvents favoring polymer-polymer interactions should be avoided, since aggregates and other macromolecular structures formed upon this type of interaction could lead to similar diffusion times for distinct polysaccharides.

β-Glucans are described to adopt a random coil conformation in DMSO, meaning that the number of polymer-polymer and polymer-solvent interactions are similar. Differently, in water there is an increase of the polymer-polymer interactions leading to the formation of triple-helix and aggregates (;). These structures would behave as single molecules in DOSY.

Beside the solvent, the formation of helix, triple-helix and aggregates depend also on the polymer concentration. Conformational studies of polysaccharides are performed in dilute solutions, but it is not possible to discard the formation of ordered structures in the NMR experiment, since the concentration is higher. However, one evidence showing that at least most of the β-glucans chains are in a random coil conformation is the presence of the ^{13}C signal attributed to the C-3 of the 3-linked glucose units (Unit A and A'). The study of the transition from triple helix to coil using the β-glucans from *Lentinula edodes* showed that the obvious difference between the triple-helix and random coil conformation is that in the former condition, the C-3 signal cannot be observed in the ^{13}C NMR spectrum ( ).

D_2O/NaOD could also be used as a solvent, since β-glucans may also adopt a random coil conformation in NaOH. In this case, the order-disorder transition depends on the increase of alkali concentration and on the DP of the polysaccharide. For instance, the conformational

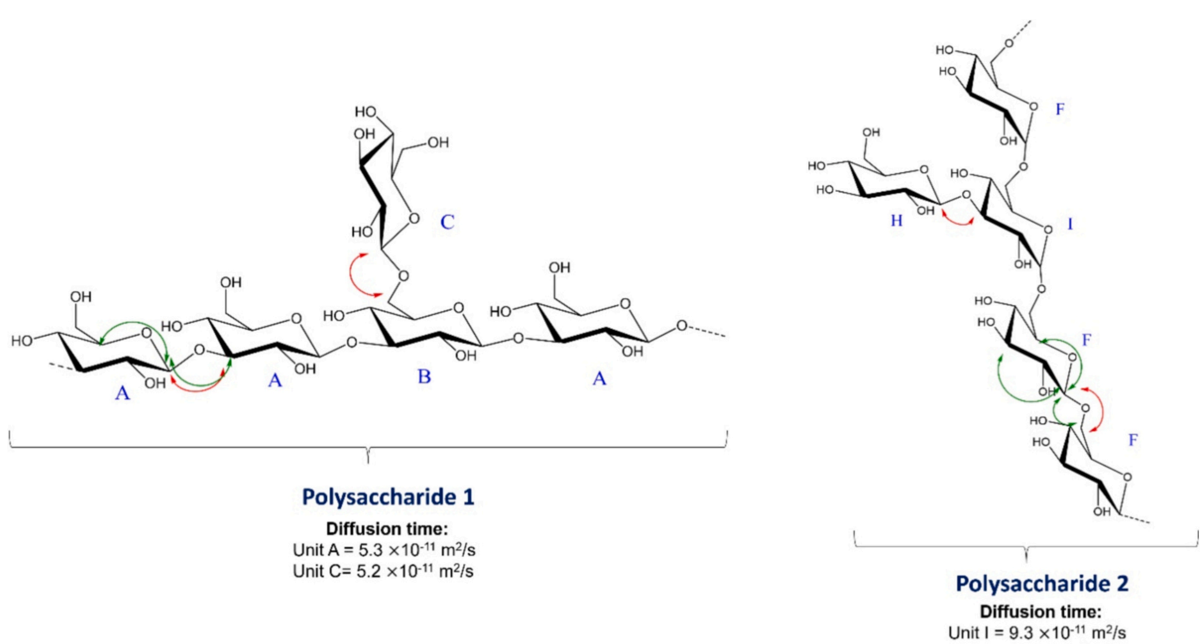

Fig. 8. Proposed chemical structures of the β-glucans present in **PEA2-SP**. The arrows represent the HMBC (red) and ROESY (green) correlations.

change of dextrans presenting a DP of 200–620 occurs at 0.19–0.24 M ( ). For β-glucans with higher DP, this transition could occur at higher NaOH concentrations since the ordered structures increases according to the DP ( ). In addition, in our previous work, the Congo red analysis suggested that not all ordered structures were disrupted at NaOH concentration of 0.5 M ( ).

DMSO was selected in the present work to obtain conditions where most of the different types of β-glucans are present as random coil, minimizing the formation of aggregates. Different situation could occur for other types of polysaccharides.

The second point is that chemical shifts presenting the same diffusion time could either belong to units of the same polysaccharide chain or to distinct polysaccharides of the same molar mass, as discussed for DOSY interpretation of **YBGS**. This may occur even when the solvent doesn't favor polymer-polymer interactions.

It should be taken into consideration that not all chemical shifts in the NMR spectrum can be used for integration and determination of the diffusion time if the aim of the experiment is to determine the different linkage patterns of similar polysaccharides. This is due to the overlapping of signals belonging to distinct units. Therefore, a complete NMR assignment of the units is necessary prior the DOSY analysis. In the present work, the chemical shifts arising from the anomeric hydrogens were used, since they are less overlapped than those belonging to H2-H6.

Much of the chemical structure elucidation of the β-glucans, as well as other types of polysaccharides, is classically performed using chemical modifications such as controlled Smith degradation and acid or enzymatic degradation (; ; ; ;). In the present work we demonstrated that DOSY may be useful for distinguishing ambiguous chemical correlations among the distinct fractions and should be included in the NMR experiments routine to predict the types of structures present in the sample in a faster manner when compared to the above-mentioned strategies. For the elucidation of the carbohydrate chemical structure, DOSY should be used as a complementary technique.

5. Conclusion

By using methylation analysis and 2D NMR techniques such as COSY, TOCSY, HMBC, H2BC, HSQC-TOCSY, NOESY and ROESY, it was possible to identify the linkages present in the three β-glucans samples studied in the present work. Although these methodologies were enough to predict the structure of a homogeneous β-glucan sample with a repetitive structure, the interpretation of samples containing mixtures of β-glucans was ambiguous until DOSY was included. The present work demonstrates the potential of the DOSY technique as an auxiliary technique for polysaccharide chemical structure determination. This work also highlights the importance of a thorough structural analysis, as samples presenting similar NMR signals at a glance appear to consist of distinctly different β-glucans after in-depth analyses. Such information is of great interest regarding the improved study of correlations between the molecular structure of polysaccharides and their respective biological activities.

CRediT authorship contribution statement

Mariana Mazetto de Carvalho: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Christiane Færestrand Ellefsen:** Writing – review & editing, Investigation, Conceptualization. **Andrea Angelov Eltvik:** Writing – review & editing, Investigation, Formal analysis. **Marianne Hiorth:** Writing – review & editing, Supervision, Conceptualization. **Anne Berit C. Samuelsen:** Writing – review & editing, Supervision, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at .

Data availability

Data will be made available on request.

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